# F-01-CX-L3-concentration-kill-test v1: does L3 divergence force stronger concentration incompatible with W+ divergence? Single kill audit

ZFL Lab

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MISSION: Single kill test for new family F. Separate explicitly $C_{L^3} = \limsup \|u\|_3 = \infty$ (PROVEN via ESS) from any additional space-time concentration claim. Search $T^* < \infty \Rightarrow C_F$ strictly stronger than $C_{L^3}$ involving scale+time+concentration that could yield $C_E^{weak} \wedge C_{L^3} \wedge C_F \Rightarrow \perp$ with $\int_0^{T^*} W_+ = \infty$.

INPUT: $\mathcal{C}_{PROVEN} = \{C_E^{weak} : \limsup_{t\uparrow T^*} \int_0^t W = \infty \Rightarrow \int_0^{T^*} W_+ = \infty,\ C_{L^3} : \limsup_{t\uparrow T^*} \|u\|_3 = \infty\}$, G locked, E-05-¿E-12 CLOSED/BLOCKED.

FORBIDDEN: Assume $C_{L^3} \Rightarrow$ concentration; need proof. $C_F \equiv C_{L^3}$. Concentration possible not necessary. CKN/epsilon-regularity restatement without new necessary divergence. Qualitative link to $W_+$ only. Saving via F-02.

AUDIT:

1. Scale exact $L^3$: Navier-Stokes scaling $u^\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, $\|u^\lambda\|_3 = \|u\|_3$ critical. ESS theorem $T_{max} < \infty \Rightarrow \limsup \|u\|_{L^3} = \infty$ for suitable weak solutions.

Does $\limsup \|u\|_3 = \infty$ imply $\exists t_n \uparrow T^*, x_n, r_n \downarrow 0$ with $\|u(t_n)\|_{L^3(B_{r_n}(x_n))} \geq \epsilon_0$? Profile decomposition (Gallagher-Koch-Planchon) gives after rescaling $\lambda_n \downarrow 0$, $\lambda_n u(\lambda_n \cdot + x_n, \lambda_n^2 \cdot + t_n) \to U \neq 0$, $\|U\|_3 \geq \epsilon_*$.

Formulate $C_F^{conc}$: $\boxed{\exists t_n \uparrow T^*, x_n, r_n \downarrow 0 : \|u(t_n)\|_{L^3(B_{r_n}(x_n))} \geq \varepsilon_0,\ r_n \sim \sqrt{T^* - t_n}}$

2. Critical space-time CKN: singular point $z_0 = (x_0, T^*)$ implies $\boxed{\limsup_{r\downarrow 0} r^{-2} \int_{Q_r(z_0)} |u|^3 + |p|^{3/2} \geq \varepsilon_0}$ call $C_F^{CKN}$.

3. Incompatibility with $\int_0^{T^*} W_+ = \infty$?

$C_E^{weak}$ says $\frac{1}{2}\|\omega(t)\|_2^2 + \nu \int_0^t \|\nabla\omega\|_2^2 = \frac{1}{2}\|\omega_0\|_2^2 + \int_0^t W$. Since $\limsup \int_0^t W = \infty$, $\int_0^{T^*} W_+ = \infty$ regardless of concentration. Concentration does not bound $W_+$ above.

Thus $\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \text{ compatible, no } \perp}$ $\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{CKN} \text{ compatible, no } \perp}$

Conclusion: $C_{L^3}$ does force some local concentration $C_F^{conc}$ and $C_F^{CKN}$, so $C_F$ is strictly stronger than $C_{L^3}$, but it is already known via profile / CKN, i.e., reproduces epsilon-regularity already known.

Classification: OPEN as $C_F^{conc/CKN}$ strictly stronger than $C_{L^3}$, but KILLED as bridge to $\perp$ since $C_E^{weak} \wedge C_{L^3} \wedge C_F \not\Rightarrow \perp$.

$\mathcal{C}_{PROVEN}$ could be extended to $\{C_E^{weak}, C_{L^3}, C_F^{conc}\}$ if we accept concentration as PROVEN, but central question $C_E^{weak} \wedge C_{L^3} \Rightarrow \perp$ remains OPEN.

Next: decide if F-01 merits F-02 or move to H-01.